

AIGR Gene Hypothesis Deep Research — GEI-17 (Q94361): SAP/SIM-Mediated Ligase-Independent Transcription- Inhibitor Activity

Target: *Caenorhabditis elegans* GEI-17 (UniProt Q94361) · **Term under evaluation:** transcription regulator inhibitor activity (GO:0140416) · **Focus:** function_assignment · **Hypothesis slug:** gei17-sap-sim-coregulator

Executive Judgment

Verdict: REFUTED as a molecular-function mechanism / OVER-ANNOTATED as a GO MF term — despite a partially correct structural premise.

The seed hypothesis makes a three-part claim: (1) GEI-17 retains the PIAS N-terminal **SAP domain**; (2) it retains **SUMO-interacting motifs (SIMs)**; and (3) these modules together confer a **ligase-independent transcriptional co-repressor / transcription-regulator-inhibitor activity** (GO:0140416) analogous to human PIAS1, operating **independently of the SP-RING SUMO ligase**. Our investigation finds that the structural premise is partly salvageable but the functional conclusion does not hold, for three convergent reasons:

1. **The SAP premise is sequence-undetectable but structurally plausible.** No sequence-profile method (Pfam PF02037, PROSITE PS50800, CATH-Gene3D, SUPERFAMILY) calls a SAP domain in GEI-17, whereas all three reference orthologs (human PIAS1, PIAS4, *Drosophila* Su(var)2-10) are annotated by multiple independent methods. However, AlphaFold shows GEI-17 retains a confidently folded, helix-hairpin-helix N-terminal module that superposes onto the PIAS1 SAP core at **1.55 Å Cα-RMSD**. So a cryptic, highly diverged SAP-like fold is present — but the **DNA-binding helix-2 and the LXXLL motif invoked by the hypothesis are degenerate** (GEI-17 "LQSII" vs PIAS1 "LQVLL").

2. **The SAP domain is the wrong module even in human PIAS1.** Decisively, the ligase-independent transcription-inhibitor activity of human PIAS1 does **not** map to the SAP domain. A SAP-domain mutant and a catalytically dead (ligase-dead) PIAS1 **both retain** IRF3 inhibition, which instead maps to a **C-terminal acidic region** (

[P 24036127](#)

The original human PIAS1 GO:0140416 basis

[P 9724754](#)

is **STAT1 sequestration** — a protein-protein mechanism, not SAP-DNA co-repression. So even if GEI-17's SAP-like fold were intact, SAP conservation would not establish GO:0140416.

3. **GEI-17's only documented worm transcription-repressive role is ligase-DEPENDENT.** GEI-17 does inhibit transcription in *C. elegans* — specifically piRNA transcription

[P 40316696](#)

— but this operates through its **SUMO E3 ligase (SP-RING) activity** and SUMOylation-driven condensate regulation, **directly contradicting** the hypothesis's "independent of its SP-RING ligase" claim.

Bottom line for the curator: GEI-17's GO:0140416 annotation is **IBA-only** (phylogenetic propagation from the PIAS/PANTHER node), with **no experimental *C. elegans* support**. The gene product's primary, experimentally-grounded molecular function is **SUMO E3 ligase activity (GO:0061665)**. Any transcriptional repression GEI-17 mediates is best captured as a **downstream biological process** (e.g., negative regulation of transcription, GO:0045892) executed *via* its catalytic ligase activity — **not** as a direct, ligase-independent transcription-regulator-inhibitor molecular function. The most important caveat is that the structural SAP-like fold is genuinely present (so a curator should not claim "SAP is absent"); the point is that its presence does not license the MF term.

Key Findings

Finding 1 — GEI-17 lacks a sequence-detectable SAP domain, unlike all three reference orthologs

InterPro integrated annotation calls the SAP domain (IPR003034) in human PIAS1 (O75925, residues 11–45), PIAS4 (Q8N2W9, 12–46) and *Drosophila* Su(var)2-10 (Q7KNF5, 2–36) by **multiple independent methods** — Pfam PF02037, PROSITE PS50800, CATH-Gene3D G3DSA:1.10.720.30, and SUPERFAMILY SSF68906. For GEI-17 (Q94361), **none** of these methods call a SAP domain; only the PINIT domain (residues 203–367) and the SP-RING/MIZ zinc-finger (400–485) are detected. GEI-17 carries a long, largely disordered N-terminal extension — PINIT begins at residue ~203 versus ~119–124 in the orthologs.

Direct Smith–Waterman alignment (BLOSUM62) of the PIAS1 SAP core (residues 6–46) against each N-terminus quantifies the divergence: PIAS1-vs-PIAS4 scores 133 (63% identity); PIAS1-vs-Su(var)2-10 scores 61 (41% identity); PIAS1-vs-GEI-17 scores only 36 over a 27-aa window (37% identity). A shuffle null model (n=300 shuffles) gives GEI-17 an observed $z = 3.93$, $p \approx 0.003$ — a statistically detectable but weak remnant, confined to **SAP helix-1** (GEI-17 "VMKLRVHDLQSII" vs PIAS1 "VMSLRVSELQVLL"). Critically, the **DNA-binding second helix** ("RNKHGRKHELLTKALHLLK" in PIAS1) and the **LXXLL motif** are **not** conserved (GEI-17 "LQSII" vs PIAS1 "LQVLL"). This is the first and most direct refutation of the hypothesis's sequence premise: the specific DNA-contacting residues that the seed invokes are gone.

Finding 2 — GO:0140416 is IBA-only in GEI-17; the human basis is STAT1 sequestration, not SAP co-repression

QuickGO shows GEI-17 (Q94361) carries GO:0140416 (transcription regulator inhibitor activity) **solely with evidence code IBA** (GO_REF:0000033; phylogenetic propagation from the PANTHER PIAS-family node PTN000845825, family PTHR10782). There is **no experimental (IDA/IMP/IPI)** *C. elegans* annotation for this term. The human PIAS1 GO:0140416 annotation rests on an IDA from Liu et al. 1998 ([PMID:9724754](#)): "*PIAS1, but not other PIAS proteins, blocked the DNA binding activity of Stat1 and inhibited Stat1-mediated gene activation in response to interferon.*" This is a **protein-sequestration mechanism** dependent on the PIAS1–STAT1 interaction — **not** on SAP–DNA binding or the LXXLL motif invoked by the seed. Moreover, *C. elegans* lacks a canonical interferon/JAK–STAT1 axis, so the specific documented

mechanism underlying the human annotation does not transfer to the worm. The IBA propagation therefore carries a function whose experimental anchor is both mechanistically and organismally mismatched to GEI-17.

Finding 3 — GEI-17's SIMs are experimentally characterized for SUMO-scaffolding, not transcriptional repression

Pelisch et al. 2017 (PMID:27939944) show experimentally that "*GEI-17 and another RC component, the kinase BUB-1, contain functional SUMO interaction motifs (SIMs), allowing them to recruit SUMO modified proteins, including KLP-19, into the RC [meiotic ring complex].*" Our motif scan of GEI-17 recovers acidic-flanked hydrophobic SIM-like cores (e.g., residues 472–475 and regions near 545–548 and 597–600 within C-terminal acidic/disordered stretches), consistent with SIM presence. **So the second structural premise of the hypothesis — that GEI-17 has SIMs — is correct.** However, these SIMs function in a **SUMO–SIM scaffolding network for chromosome congression during oocyte meiosis**, operating *downstream of* GEI-17's SUMO E3 ligase activity — not in transcriptional co-repression. All experimentally characterized GEI-17 roles (meiotic ring-complex assembly,

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mitotic chromosome assembly / cell-cycle progression, PMID:25475837) are SUMO-ligase-dependent. The SIMs are real but repurposed for a catalytic/scaffolding pathway, not for the ligase-independent repression the seed proposes.

Finding 4 — AlphaFold rescues the SAP-fold premise (1.55 Å RMSD) even though sequence profiles miss it

AlphaFold models show the PIAS SAP region is confidently folded in all orthologs (PIAS1 res 11–45 mean pLDDT 93.7; PIAS4 12–46, 93.5; Su(var)2-10 2–36, 89.0). GEI-17 (Q94361) likewise has a confidently folded N-terminal module at res ~21–80 (mean pLDDT ~86–89; 100% of res 21–60 above pLDDT 70), flanked by a long disordered linker (res 81–200, mean pLDDT ~36) before PINIT (203+). CA(i,i+4) helix mapping shows GEI-17 res 20–85 forms helical runs [7,11,8,10] and PIAS1 res 6–50 forms [4,11,12] — both consistent with a SAP **helix-hairpin-helix** core. Kabsch superposition of the sequence-aligned Ca atoms (PIAS1 SAP residues 11–36 ↔ GEI-17 27–47, 26 residue pairs) gives **Ca-RMSD = 1.55 Å**.

This is the key nuance for the curator: GEI-17 **retains a structurally SAP-like folded module** even though Pfam/PROSITE/CATH/SUPERFAMILY do not classify it as SAP, because the sequence has diverged beyond profile-detection limits. The structural premise of the hypothesis is

therefore **partly vindicated** — but with the same retained caveat: the DNA-binding helix-2 and the LXXLL motif are not conserved. A fold without its functional residues does not recover the function.

Finding 5 — Human PIAS ligase-independent transcription inhibition is SAP-INDEPENDENT, undercutting the seed's core mechanism

This is the decisive functional finding. Li et al. 2013 ([PMID:24036127](#)) dissected exactly the ligase-independent transcription-inhibitor activity the seed invokes and found it is **neither ligase-dependent nor SAP-dependent**:

- **Ligase-independent:** *"SUMO E3 ligase activity dead mutant PIAS1/C350S still had the comparable inhibitory function with WT PIAS1."*
- **SAP-independent:** *"PIAS1 with a mutation in the SAP domain retained the inhibitory function in virus-induced IFN transcription."*
- **The responsible module is elsewhere:** *"The C-terminal region of PIAS1 around a cluster of acidic amino acids is critical for the interaction with IRF3 and the inhibitory functions of PIAS1."*

Consistent with this, Tolkunova et al. 2007 ([PMID:17991485](#)) showed PIASy inhibits Oct4-mediated transcription by sequestering Oct4 to the nuclear periphery, and *"These modes of PIASy action are uncoupled from its sumoylation activity."* In other words, even where PIAS proteins do exhibit genuine ligase-independent transcriptional inhibition, the mechanism is **transcription-factor protein-protein sequestration via C-terminal/acidic regions**, not SAP-domain DNA binding or the LXXLL motif. The seed hypothesis's specific causal claim — that the SAP domain confers the activity — is directly contradicted in the very paralog it uses as its model.

Finding 6 — GEI-17 does inhibit transcription in *C. elegans*, but via ligase-DEPENDENT condensate regulation

Zhu et al. 2025 ([PMID:40316696](#)) provide the direct worm evidence that GEI-17 represses transcription — and show it is ligase-dependent. A screen for regulators of piRNA transcription *"isolated the SUMO E3 ligase GEI-17 as inhibiting and the SUMO protease TOFU-3 as promoting piRNA transcription foci formation, thereby regulating piRNA production."* The effect operates through **SUMOylation/deSUMOylation of the USTC transcription complex and phase-separation of transcriptional condensates** — i.e., GEI-17's SUMO E3 ligase (SP-RING) activity produces the transcription-inhibitory outcome (deSUMOylation by TOFU-3 *promotes* foci;

SUMOylation by GEI-17 opposes them). This is the opposite of the hypothesis's "independent of its SP-RING ligase" claim. Reinforcing a ligase-dependent, transcription-factor-SUMOylation mode, Fergin et al. 2022 ([PMID:35666766](#)) show GEI-17-dependent "*sumoylation of the ETS transcription factor LIN-1 at K169 is necessary for the proper contraction of the ventral vulA toroids*" during vulval development — GEI-17 acting on a transcription factor by catalytic SUMOylation, not by ligase-independent co-repression.

Mechanistic Model / Interpretation

The investigation can be summarized as a decision tree in which the hypothesis is tested at three independent gates, failing the functional gates even where it passes a structural one:

SEED HYPOTHESIS: GEI-17 SAP + SIMs → ligase-INDEPENDENT
transcription-regulator-inhibitor activity (G0:0140416)

GATE 1: Does GEI-17 retain the SAP domain?

Sequence profiles (Pfam/PROSITE/CATH/SUPERFAMILY):	NO	F001
AlphaFold fold superposition (1.55 Å RMSD):	PARTIAL	F004
DNA-binding helix-2 + LXXLL residues:	NO	F001/F004
→ Fold cryptically present; functional residues GONE		

(even granting the fold...)

GATE 2: Does the SAP domain confer the activity
in the model paralog (human PIAS1)?

PIAS1 ΔSAP mutant RETAINS IRF3 inhibition:	NO	F005
PIAS1 ligase-dead (C350S) RETAINS inhibition:	NO	F005
Activity maps to C-TERMINAL ACIDIC region:	(not SAP)	F005
Original G0 basis = STAT1 SEQUESTRATION:	(not SAP)	F002
→ SAP is the WRONG module; mechanism is TF seq.		

(so is GEI-17's worm repression ligase-independent?)

GATE 3: In *C. elegans*, is GEI-17 repression
independent of its SP-RING ligase?

piRNA transcription repression = SUMO-ligase DEPENDENT	F006
LIN-1 regulation = catalytic SUMOylation (K169):	DEP. F006
SIMs = SUMO scaffolding of meiotic RC (ligase pathway)	F003
→ Repression is a DOWNSTREAM BP of catalytic activity	

CONCLUSION: G0:0140416 (MF) NOT supported for GEI-17.

Primary MF = SUMO E3 ligase (G0:0061665).

Transcription repression = downstream BP (e.g., G0:0045892),
executed VIA ligase activity.

The coherent narrative: GEI-17 is, first and foremost, the *C. elegans* PIAS-family **SUMO E3 ligase**. Its experimentally documented biology — meiotic ring-complex assembly, mitotic chromosome assembly, telomere anchoring, vulval LIN-1 regulation, and piRNA condensate control — is uniformly **ligase-dependent**. The seed hypothesis attempts to graft onto GEI-17 a *ligase-independent* co-repressor function borrowed from human PIAS1, routed through the SAP domain and LXXLL motif. But that route is a misreading of the human data: even in PIAS1 the ligase-independent transcription-inhibitor activity is SAP- and ligase-independent, mapping to

a C-terminal acidic TF-sequestration module. GEI-17 retains a cryptic SAP-like *fold* but not its DNA-binding residues, and its worm repression runs entirely through catalysis. The GO:0140416 annotation is thus an IBA artifact of phylogenetic propagation from a family node whose experimental anchor (human STAT1/IRF3 inhibition) neither depends on the conserved module nor exists in the worm's signaling repertoire.

Evidence Matrix

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence / limitations
InterPro / Pfam / PROSITE / CATH / SUPERFAMILY	Structural/ evolutionary; computational	Refutes sequence premise	Does GEI-17 retain a SAP domain by profile methods?	No SAP (IPR003034) called in GEI-17 by any method; called in all 3 orthologs by ≥ 4 methods. SW remnant weak ($z=3.93$, $p \approx 0.003$), helix-1 only; helix-2 + LXXLL not conserved	Protein sequence; cross-ortholog	High for "profile it"; can exclude cryptic (see AlphaFold)
AlphaFold models	Structural/ evolutionary; computational	Qualifies (partially supports structural premise)	Does GEI-17 retain a SAP-like 3D fold?	GEI-17 N-terminal module folds (pLDDT ~86–89); superposes on PIAS1 SAP at 1.55 Å Ca-RMSD over 26 residues	Predicted structure	Model high; prediction not experimental DNA-binding residue degenerate
9724754	Direct assay (IDA)	Qualifies/ Refutes mechanism	Experimental basis of human PIAS1 GO:0140416?	PIAS1 blocks STAT1 DNA binding and IFN-induced gene activation — sequestration , not SAP co-repression	Human cells, IFN signaling	High; mechanism TF sequences and weak lacks JAK2/STAT1

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence / limitations
24036127	Direct assay (mutant mapping)	Refutes seed's core mechanism	Does SAP/ligase confer PIAS1 transcription inhibition?	ΔSAP mutant AND ligase-dead C350S both retain IRF3 inhibition; activity maps to C-terminal acidic region	Human cells, virus-induced IFN	High; consistent against media model
17991485	Direct assay	Refutes SAP mechanism (supports alt)	Is PIAS ligase-independent repression SAP-mediated?	PIASy represses Oct4 by nuclear-periphery sequestration; <i>"uncoupled from its sumoylation activity"</i>	Human/mouse pluripotency	High; independent paralogs conclude (TF seques
27939944	Direct assay / interaction	Qualifies (SIMs real; wrong function)	Does GEI-17 have functional SIMs, and for what?	GEI-17 SIMs recruit SUMO-modified KLP-19 into the meiotic ring complex	<i>C. elegans</i> oocyte meiosis	High; S confirm function SUMO scaffold ligase-depend
40316696	Mutant phenotype / screen	Refutes "ligase-independent"; supports BP	Does GEI-17 repress worm transcription, and how?	GEI-17 inhibits piRNA transcription foci via SUMOylation of USTC / condensate control (ligase-dependent)	<i>C. elegans</i> germline, piRNA	High; consistent worm transcr evidence ligase-DEPEND

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence / limitations
35666766	Mutant phenotype	Supports catalytic model	Does GEI-17 act on TFs by SUMOylation?	GEI-17-dependent SUMOylation of ETS TF LIN-1 at K169 required for vulA toroid contraction	<i>C. elegans</i> vulval development	High; catalytic model (ligase dependent) TF regulation
25475837	Mutant phenotype	Supports primary function	What is GEI-17's core cellular role?	PIAS(GEI-17) SUMO E3 ligase required for mitotic chromosome alignment / cell-cycle progression	<i>C. elegans</i> embryo	High; reinforcement of SUMOylation as primary function

GO Curation Implications

Lead for curator verification — likely action: do NOT retain GO:0140416 as a direct molecular-function annotation for GEI-17; treat as non-core / downstream.

- **GO:0140416 (transcription regulator inhibitor activity, MF):** The evidence does **not** support this as a direct molecular function of GEI-17. It is **IBA-only**, propagated from the PIAS/PANTHER family node, whose experimental anchor (human PIAS1 STAT1/IRF3 inhibition) (i) is mechanistically a TF-sequestration activity mapping to a C-terminal acidic region rather than the conserved SAP domain, and (ii) does not depend on the module the seed invokes. Recommended action: **remove or demote** this MF annotation for GEI-17, or at minimum flag it as an uninformative IBA carry-over not supported by the worm's experimental record. If retained for phylogenetic-consistency reasons, it should carry a NOT/under-review qualifier pending experimental validation.

• **Primary molecular function — GO:0061665 (SUMO ligase activity) / SUMO transferase:**

This is the experimentally supported MF and should be the anchor annotation

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P 27939944;
P 35666766;
P 40316696)

all rely on GEI-17 catalytic activity).

• **Biological process — GO:0045892 (negative regulation of DNA-templated transcription) or a piRNA-transcription-specific child:** GEI-17's documented transcription-repressive effect on piRNA genes

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is real and supportable as a **BP**, but it is executed *via* SUMO ligase activity and condensate regulation — a downstream process, **not** a direct MF. A curator could add a BP term with IMP evidence from

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explicitly noting ligase dependence.

• **Avoid "protein binding" as a terminal recommendation:** The more informative supported terms are SUMO ligase activity (MF) and negative regulation of transcription (BP, ligase-dependent); these should be used rather than the uninformative generic binding term.

Mechanistic Scope

Immediate molecular activity being tested: a *ligase-independent* transcription-regulator-inhibitor (co-repressor) activity attributed to the SAP domain + SIMs. **This direct activity is not supported.** What is directly supported is **SUMO E3 ligase (transferase) activity** (SP-RING/MIZ domain, res 400–485), aided by PINIT (203–367) substrate positioning and C-terminal SIMs for SUMO-chain/scaffold engagement.

Separation of direct activity from downstream effects: - *Direct gene-product activity:* catalytic SUMO conjugation onto substrate lysines (e.g., LIN-1 K169); SIM-mediated recruitment of SUMO-modified partners (KLP-19) into higher-order assemblies. - *Pathway/cellular consequence:* SUMOylation of the USTC transcription complex modulates phase-separated transcriptional condensates, thereby repressing piRNA transcription — a **downstream BP**, not a direct MF of binding/inhibiting the transcription machinery. - *Developmental/phenotypic outcome:* meiotic chromosome congression, mitotic chromosome assembly, telomere anchoring, vulval toroid contraction — pleiotropic phenotypes of the catalytic ligase, not evidence for a distinct co-repressor MF. - *Inferred-from-loss-of-function:* the piRNA screen phenotype

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demonstrates repression but attributes it to ligase activity; no loss-of-function data isolate a ligase-independent SAP-driven repression.

Conflicts and Alternatives

1. **Paralog over-generalization / frequency bias.** The hypothesis extrapolates from "human PIAS1 has GO:0140416" to "GEI-17 has it," but the human activity (a) does not require the domain

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and (b) originally rests on STAT1 sequestration

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an axis absent in *C. elegans*. This is a classic IBA over-annotation pattern.

2. **Fold vs. function mismatch.** AlphaFold shows a conserved SAP-like fold (1.55 Å RMSD), which could tempt a curator toward the MF. But the DNA-binding helix-2 and LXXLL residues are degenerate, and the fold is not the functional determinant even in PIAS1. Structural conservation here is a red herring for GO:0140416.

3. **SIMs real but repurposed.** The hypothesis correctly predicts SIMs (P 27939944), but their documented role is SUMO-scaffolding of the meiotic ring complex — a ligase-pathway function — not transcriptional co-repression. Correct prediction, wrong functional attribution.
4. **Organism-specific mechanism.** *C. elegans* GEI-17 transcription repression (piRNA; P 40316696) is ligase-dependent, directly contradicting the seed's "independent of its SP-RING ligase." The worm mechanism is catalytic SUMOylation + condensate control, not DNA-binding co-repression.
5. **Alternative interpretation favored by the data:** GO:0140416's phenotypic signal in GEI-17 is fully explained as a **downstream consequence of SUMO ligase activity**, requiring no separate co-repressor MF.
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Limitations and Knowledge Gaps

Gap	What was checked	Why it matters for curation	What would resolve it
No experimental test of a GEI-17 ligase-dead mutant for transcription repression	Literature + QuickGO; found only ligase-dependent assays	Directly adjudicates the seed's "ligase-independent" claim in the native organism	A GEI-17 SP-RING catalytic-dead (e.g., analogous to PIAS1 C350S) rescue in the piRNA / LIN-1 assays
AlphaFold fold not experimentally confirmed, and no DNA-binding assay for GEI-17 N-terminus	AlphaFold superposition (1.55 Å) + motif scan	Determines whether the cryptic SAP-like fold has any residual DNA-binding/ chromatin activity	EMSA / chromatin-binding assay of the GEI-17 N-terminal module; cryo-EM/X-ray if feasible
SIM functional repertoire beyond meiosis untested for transcription	Motif scan + P 27939944	Whether SIMs contribute to any repression independent of catalysis	SIM-point-mutant GEI-17 in transcription assays with ligase intact vs. dead
Whether GEI-17 physically contacts worm transcription factors independent of SUMOylation	Literature only	A C-terminal-acidic-style sequestration (as in PIAS1/PIASy) could exist in worm	Co-IP / proximity labeling of GEI-17 with germline TFs ± SUMOylation-site mutations
Exact boundaries/ identity of GEI-17 "acidic" C-terminal region vs. PIAS1's IRF3-binding module	Partial (SIM scan found acidic/ disordered C-terminal regions)	PIAS1's activity maps here; a worm homolog region could in principle sequester a TF	Deletion mapping of GEI-17 C-terminus in a repression assay

Discriminating Tests

1. **Ligase-dead rescue in the native assay (highest value).** Express catalytically dead GEI-17 (SP-RING Cys mutant) in a *gei-17* loss-of-function background and test rescue of piRNA transcription repression

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 assay) and LIN-1-dependent vulval phenotypes
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If repression requires catalysis, the ligase-independent hypothesis is refuted *in vivo*; if a dead mutant still represses, the hypothesis gains support.

2. **SAP-module deletion/point mutants.** Delete the AlphaFold-defined N-terminal SAP-like module (res ~21–80) or mutate its residual helix-1 residues; test for any loss of transcription repression separable from catalysis. Parallels the decisive PIAS1 Δ SAP experiment

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3. **DNA/chromatin-binding assay of the isolated N-terminal module.** EMSA or chromatin pulldown to test whether the cryptic SAP-like fold binds DNA at all, given helix-2/LXXLL degeneracy.
4. **C-terminal acidic-region mapping.** By analogy to PIAS1's IRF3-binding region, test whether a GEI-17 C-terminal fragment sequesters any germline transcription factor independent of SUMOylation.
5. **Comparative orthology refinement.** Formal structure-guided alignment (e.g., Foldseek/DALI) of GEI-17 vs. PIAS1/PIAS4/Su(var)2-10 SAP folds to quantify which functional SAP residues are retained vs. lost — strengthens the "fold-present, function-absent" curation note.

Curation Leads (require curator verification)

Candidate action change: - Remove / demote GO:0140416 (transcription regulator inhibitor activity, MF) as a direct annotation for GEI-17; it is IBA-only with no worm experimental support and rests on a human mechanism (SAP- and ligase-independent TF sequestration) that neither transfers to the worm nor depends on the conserved module. -

Retain/strengthen GO:0061665 (SUMO ligase activity, MF) as the primary molecular function. - **Consider adding** GO:0045892 (negative regulation of DNA-templated transcription, BP) with IMP evidence from

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annotated as ligase-dependent (downstream of catalytic activity).

Candidate references with exact snippets to verify: - [PMID:24036127](#): "*PIAS1 with a mutation in the SAP domain retained the inhibitory function in virus-induced IFN transcription*" and "*SUMO E3 ligase activity dead mutant PIAS1/C350S still had the comparable inhibitory function with WT PIAS1.*" → SAP- and ligase-independence of the human activity. - [PMID:9724754](#): "*PIAS1, but not other PIAS proteins, blocked the DNA binding activity of Stat1 and inhibited Stat1-mediated gene activation in response to interferon.*" → Human GO:0140416 basis is STAT1 sequestration. - [PMID:40316696](#): "*isolated the SUMO E3 ligase GEI-17 as inhibiting ... piRNA transcription foci formation.*" → Worm repression is ligase-dependent. - [PMID:27939944](#): "*GEI-17 ... contain functional SUMO interaction motifs (SIMs), allowing them to recruit SUMO modified proteins ... into the RC.*" → SIMs real, function is SUMO scaffolding. - [PMID:35666766](#): "*sumoylation of the ETS transcription factor LIN-1 at K169 is necessary for the proper contraction of the ventral vulA toroids.*" → Catalytic TF regulation.

Suggested curator questions: - Is the GEI-17 GO:0140416 IBA propagation appropriate given the family node's experimental anchor is SAP- and ligase-independent and organism-mismatched? - Should the worm piRNA repression phenotype be annotated as a BP with explicit ligase dependence rather than inherited as an MF?

Suggested experiments: ligase-dead rescue (test #1 above) and SAP-module mutant (test #2) are the two most decisive and should be flagged as the experiments that would settle the annotation.

Evidence Base Summary

The strongest primary-literature pillars, in order of decisiveness for this hypothesis:

1. [PMID:24036127](#) (Li et al. 2013) — *decisive refutation of the SAP mechanism*: human PIAS1 ΔSAP and ligase-dead mutants both retain transcription inhibition; activity maps to the C-terminal acidic region.

2. [PMID:40316696](#) (Zhu et al. 2025) — *decisive worm evidence against ligase-independence*: GEI-17 represses piRNA transcription via SUMO ligase activity and condensate regulation.
3. [PMID:9724754](#) (Liu et al. 1998) — the human GO:0140416 IDA basis is STAT1 sequestration, not SAP co-repression.
4. [PMID:17991485](#) (Tolkunova et al. 2007) — independent paralog confirms ligase-independent PIAS repression works by TF sequestration.
5. [PMID:27939944](#) (Pelisch et al. 2017) — GEI-17 SIMs are real but function in SUMO scaffolding of the meiotic ring complex.
6. [PMID:35666766](#) (Fergin et al. 2022) & [PMID:25475837](#) — reinforce that GEI-17's TF/chromosome roles are catalytic SUMO-ligase functions.

Database/computational evidence (InterPro profile methods; AlphaFold superposition) establishes the "fold present, functional residues absent" structural picture: GEI-17 escapes SAP profile detection yet retains a 1.55 Å-RMSD SAP-like fold with a degenerate DNA-binding helix-2/LXXLL motif.

Conclusion

The seed hypothesis is **refuted as a molecular-function mechanism and over-annotated as GO:0140416**, with a partially correct structural premise that must be stated carefully. GEI-17 does retain a cryptic, structurally SAP-like fold and genuine SIMs — but its DNA-binding residues are degenerate, the SAP domain is not the module that confers transcription-inhibitor activity even in human PIAS1, and GEI-17's only documented worm transcription-repressive role is SUMO-ligase-dependent. The curator should treat GO:0140416 as an unsupported IBA carry-over, anchor GEI-17's annotation on SUMO E3 ligase activity (GO:0061665), and, if desired, capture its transcription repression as a downstream, ligase-dependent biological process.